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OpenHarmony Technical Conference · Prague 2026

Bringing Oniro to a European developer phone

How OpenHarmony reaches new hardware

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Why this work

Three roads onto a device. We take the one that's open.

A Open drivers
the fully open road

Mainline Linux with open graphics and audio drivers. OpenHarmony already runs this way on the OnePlus 6T.

The cleanest stack there is — but only where community driver work has reached the chip. Rare on recent phones.

B The maker supplies it
the first-class road

The chip or device maker ships hardware support built for OpenHarmony. This is how certified products and reference boards ship.

The best possible outcome. It needs the manufacturer in the room.

C Bridge existing drivers
the road we take

Use the driver package the chip vendor already supplies, with one translation layer underneath OpenHarmony.

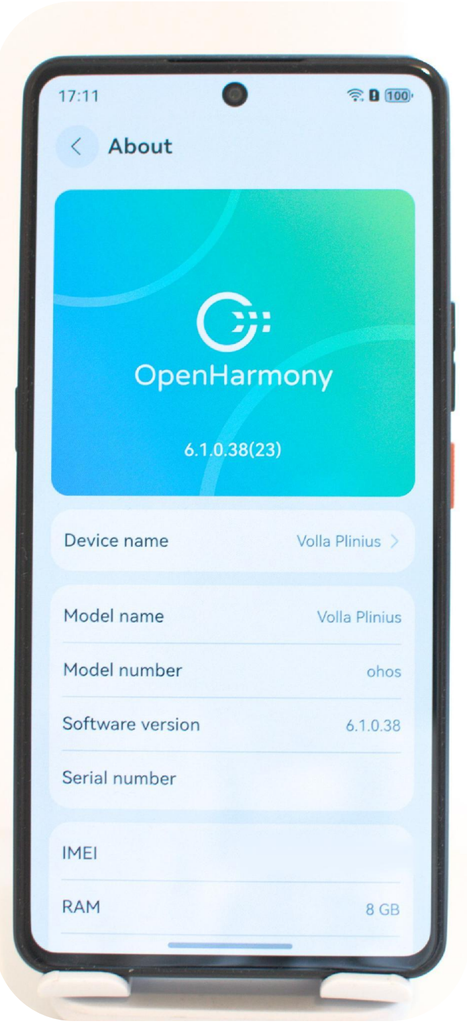
Works on hardware you can buy today. The vendor code stays a black box, and we accept that.

The target

Volla Phone Plinius

A current phone from Volla, a European maker , sold as an open device: the bootloader unlocks, and it already ships with a choice of operating systems.





Chipset	MediaTek Dimensity 7300 · Mali-G615
Display	6.67" OLED · 1080×2400 · up to 120 Hz
Battery	5,300 mAh, replaceable

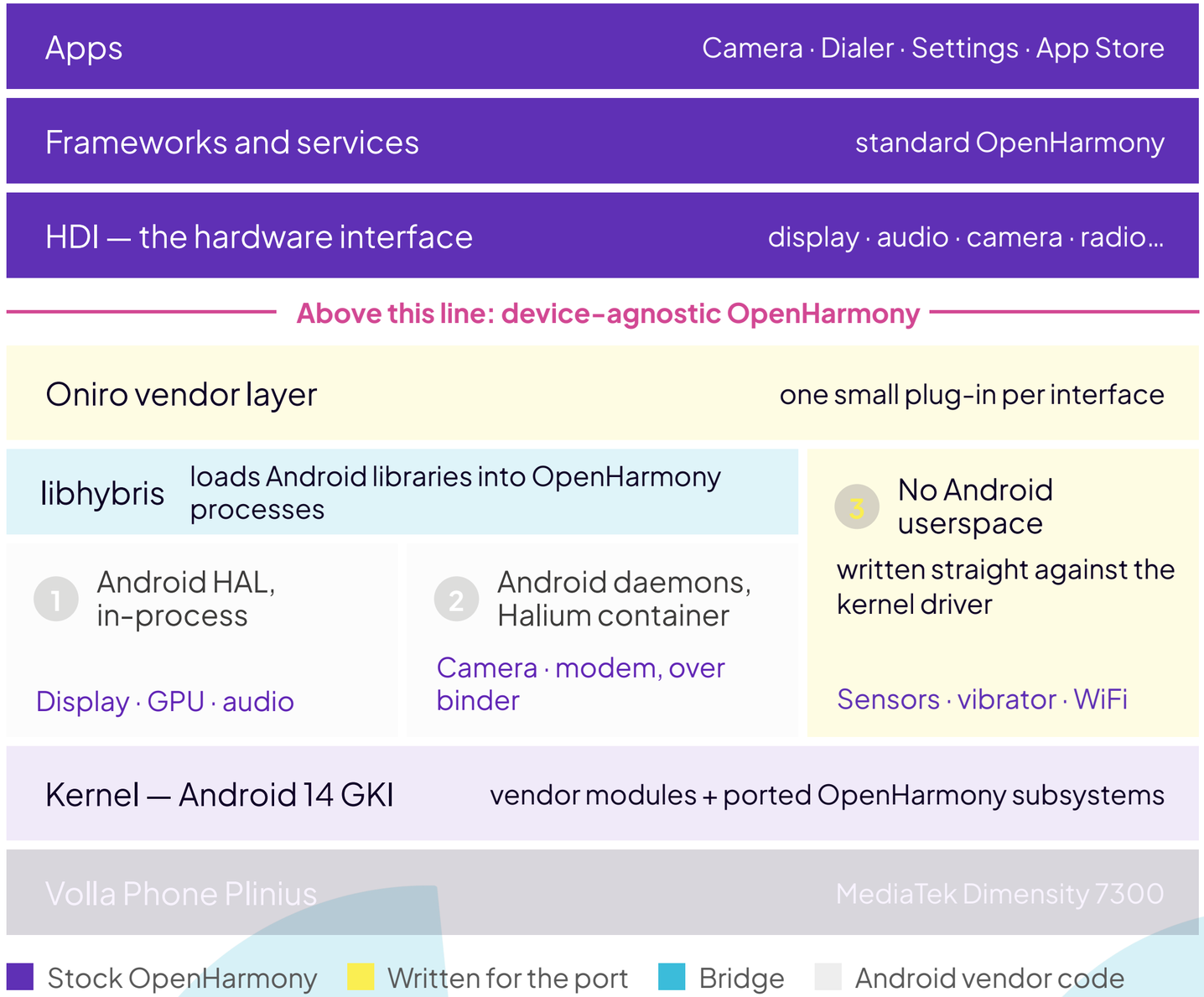
01 · How it works

OpenHarmony itself is untouched

The apps are ordinary ArkTS apps. None of them knows anything about this phone.

Frameworks and system services are the standard OpenHarmony stack, as shipped.

Beneath them sits the HDI , OpenHarmony's own hardware interface. Everything above it is device-agnostic by design.

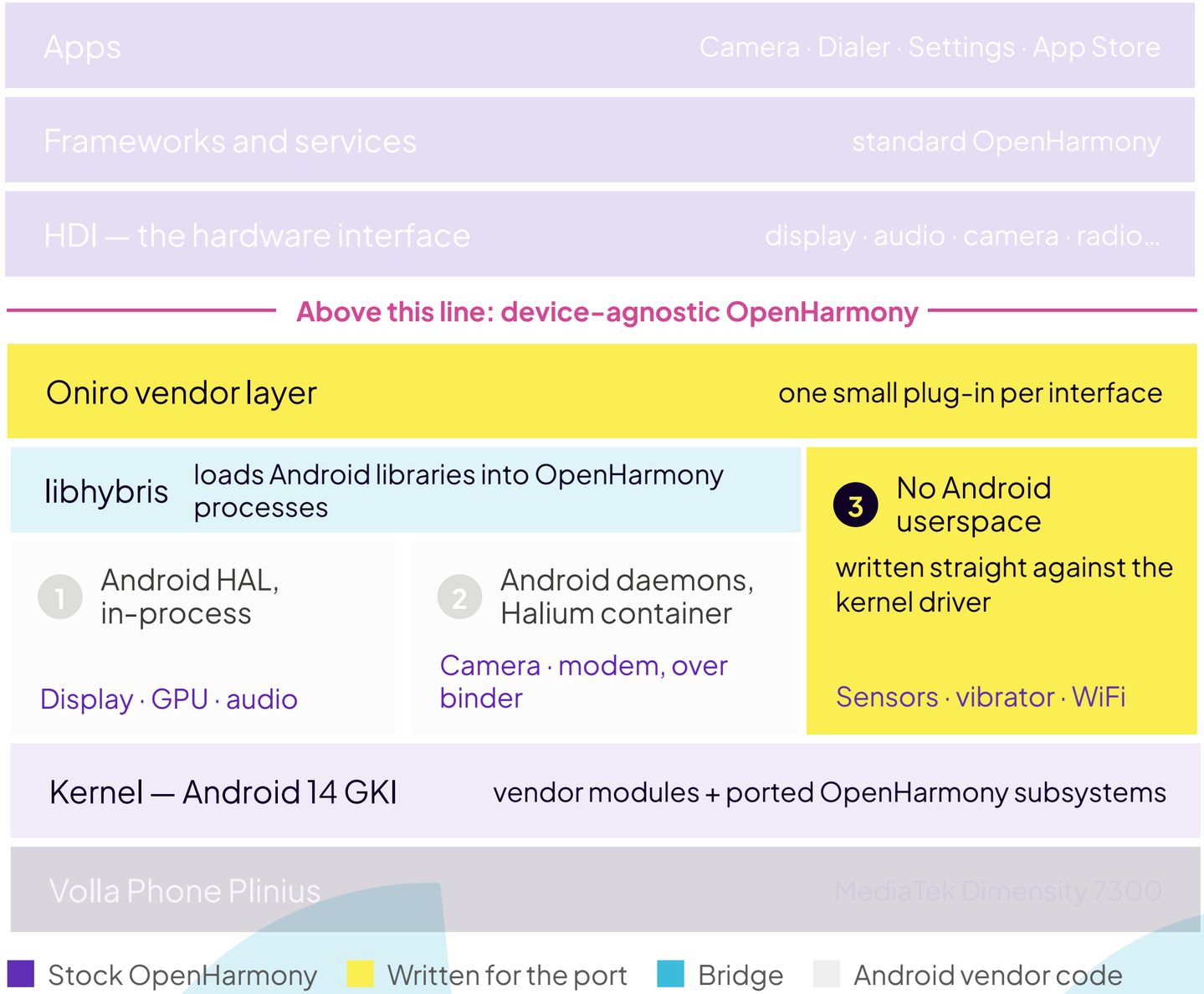


01 · How it works

One layer, three routes to the hardware

For each hardware interface, one small plug-in turns the OpenHarmony call into the vendor's equivalent. The yellow parts are the entire port.

"Everything above it is ordinary OpenHarmony. It has no idea what hardware it is talking to."

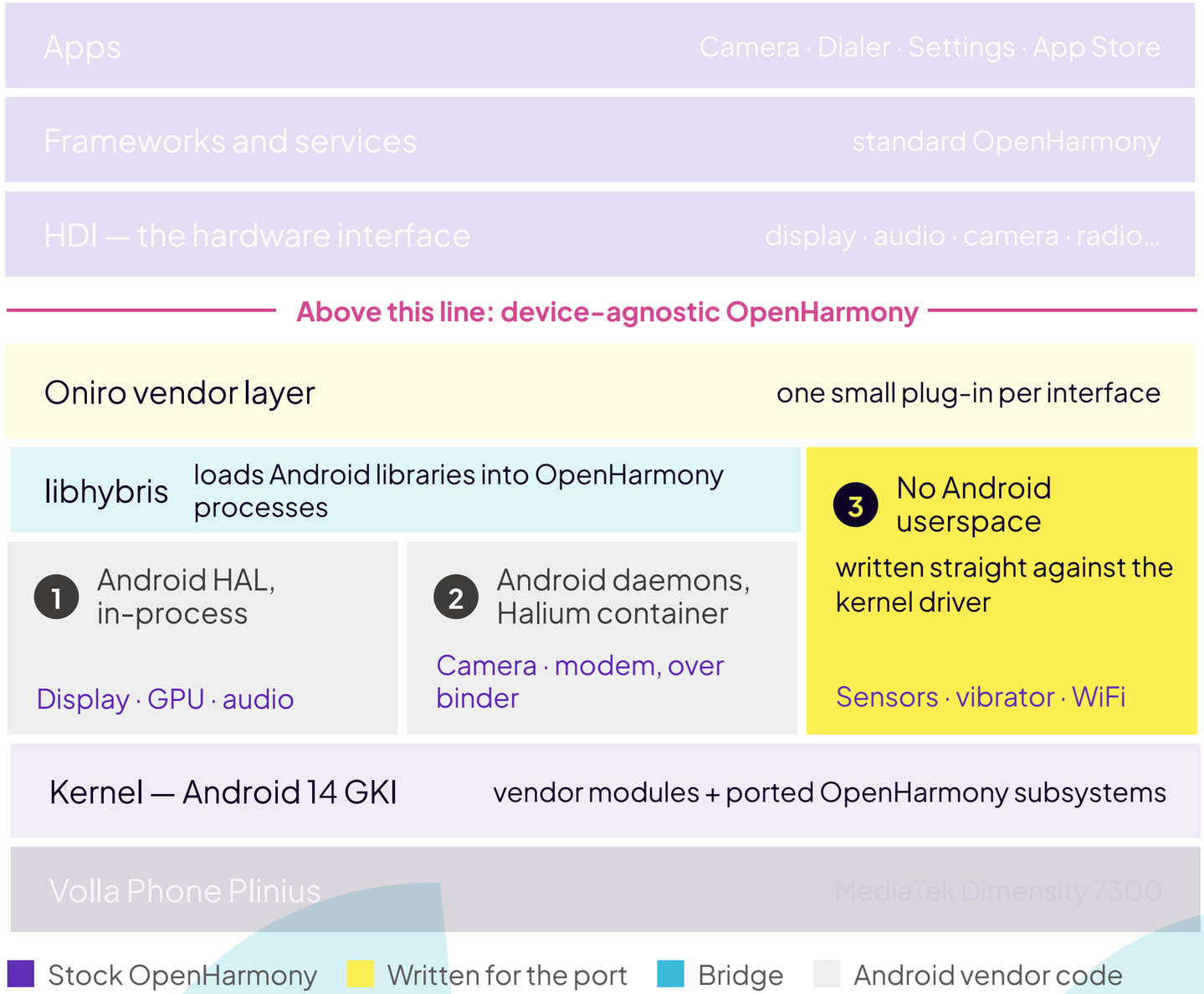


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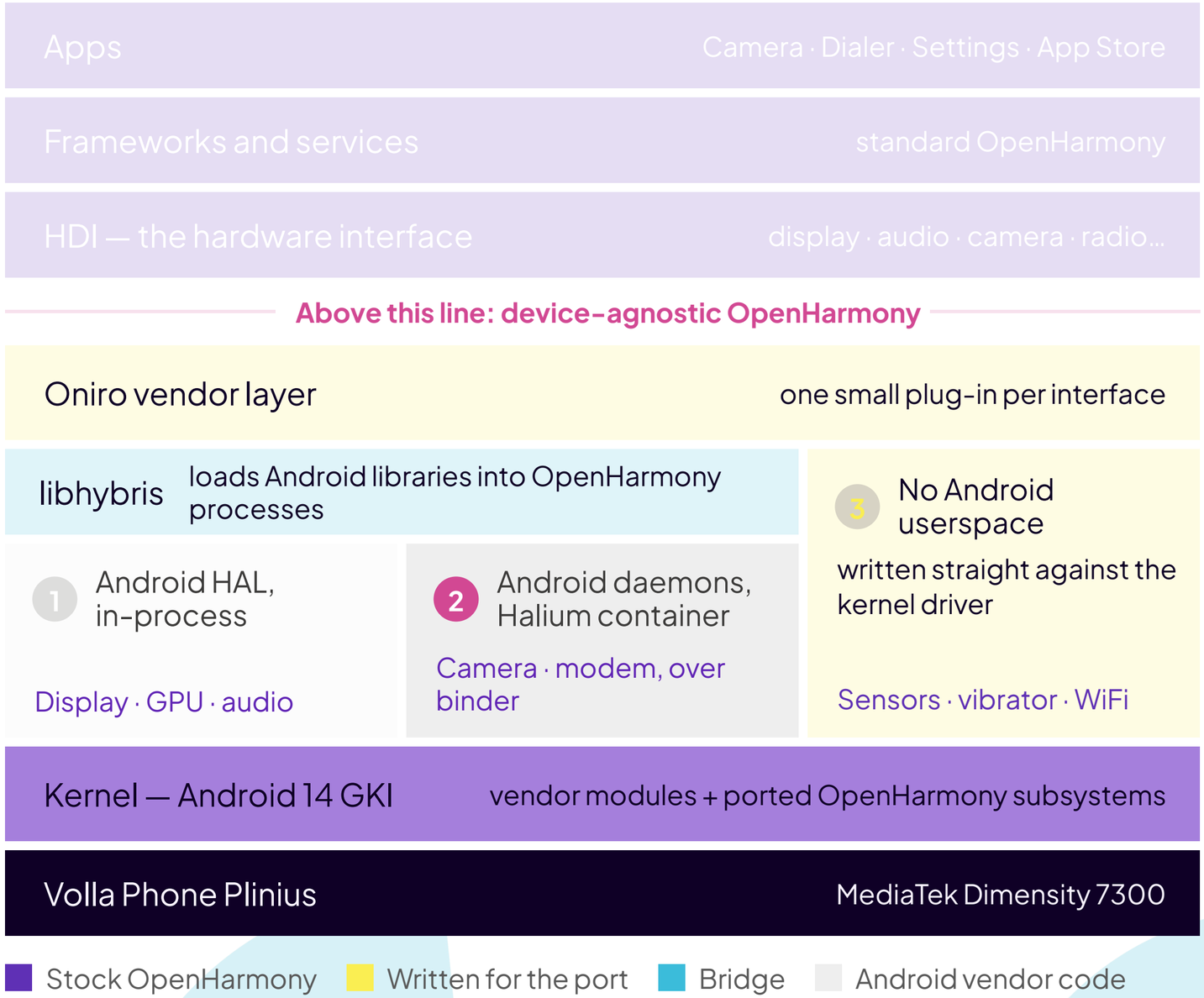
A borrowed kernel, a vendor container, almost no overhead

The kernel is the phone's own — the one the vendor drivers are built against.

A Halium container runs only the Android vendor services. No interface, no apps.

OpenHarmony starts first and owns the device.

Not emulation: native code on both sides	Graphics shared, never copied
The same hop the phone makes anyway	GPU at full frame rate



01 · How it works

The second device reused the first one's architecture

01

Volla X23

The first port. Where the hard architectural decisions were made and paid for.

02

Volla Phone Plinius

Inherited the boot approach, the Halium container, the vendor layer and the graphics and audio plug-ins. The work went into what was new about this chip.

03

The next device

Inherits it too: the reusable parts live upstream, in the Oniro board support package.

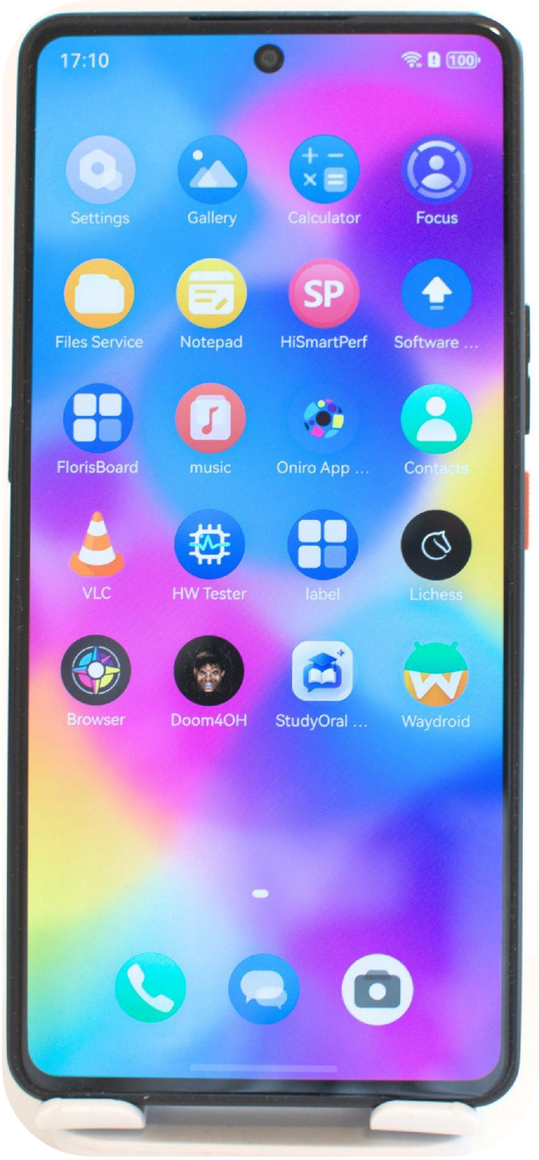




02 · Status

A phone you can actually use

Screen and touch	full Oniro interface, sized for this panel	Works
WiFi	joins a network, real traffic	Works
Audio	speaker and microphone	Works
Cameras	front and rear, standard Camera app	Works
Phone	voice calls, SMS, mobile data	Works
Sensors	plus vibration and brightness	Works
Apps	App Store, browser, media, games	Works
Android apps	optional, in their own windows	Works
Bluetooth · location · NFC · fingerprint · video recording		In progress

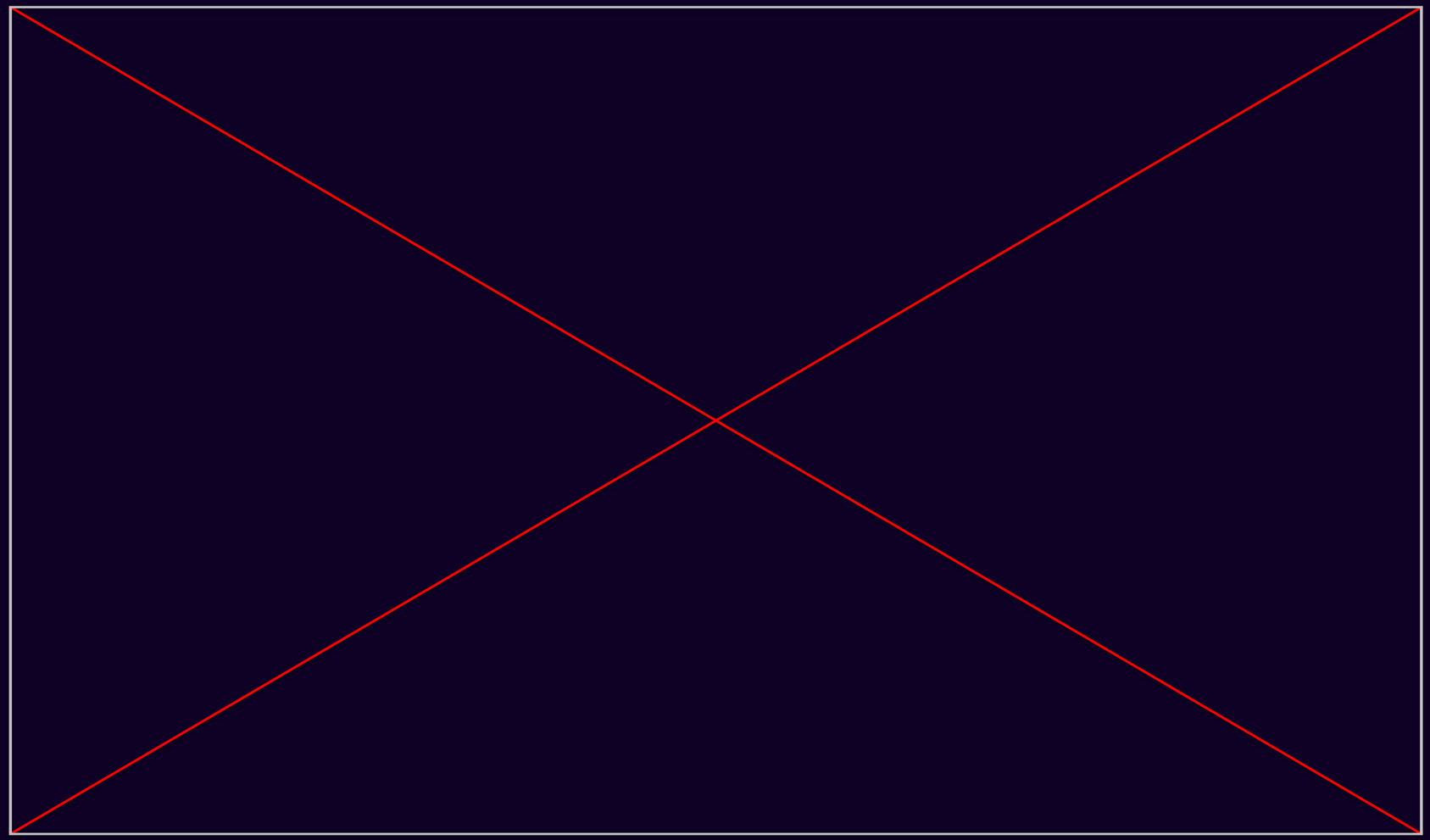


02 · Status

See it run

- Boot into Oniro
- WiFi
- Apps from the Oniro App Store
- Rear and front camera
- A real call, SMS, 4G data





02 · Status

And the apps you still depend on

Android apps run in an optional container with **Waydroid**, started only when you open one.

Each app is an ordinary OpenHarmony window, with OpenHarmony's status bar, navigation and audio.



03 · What is left

Daily-drivable for a developer. Not yet a product.

What is missing

Bluetooth — no earbuds, no car. Wireless audio is otherwise ready.

Location — the GPS runs, but isn't connected to OpenHarmony yet. No navigation.

Video recording and hardware video playback. Photos work.

NFC and fingerprint — the furthest from done.

Voice over LTE — calls use the older fallback path today.

Before you'd hand it to someone

Security features are off for development. Secure boot is intact, so turning them on is configuration, not redesign.

No over-the-air updates yet.

Battery life not tuned yet.

03 · What is left

Where this goes

Finish the connections	Bluetooth and wireless audio, location, NFC, fingerprint
Video	Recording, and hardware-accelerated playback
Telephony	Dual SIM and Voice over LTE
Make it a product	Security features on, updates, power tuning
Keep it upstream	Reusable parts go into the Oniro board support package

This is not really about one phone

2 devices

2 chipset generations

1 build



Already spreading: the community has brought up a Teracube 2s on the same layer.

Almost all commercial mobile silicon ships its drivers in the shape this layer is built to bridge.

A new device needs two things, checkable in advance: an unlockable bootloader and a vendor driver package.

All of it is open: source, build instructions, flashing steps and plans.

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Thank you

Bring us a device

Have a device in mind? The path is documented and the code is open.

Make hardware? Road B is still the best road. Let's talk.

Want to try it? Video, build and flashing steps are public.

Code	github.com/eclipse-oniro4openharmony
Docs	docs.oniroproject.org
Project	oniroproject.org
Me	Francesco Pham · LinkedIn



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Thank you

Francesco Pham · Eclipse Oniro Project Lead

